B.E. Even Semester Midterm Examination, 2023-24(Even)

Subject: Computer Organization & Architecture | **Paper Code:** PCC-IT 402

Q1. 8k memory, start 0000. Find ending address. Calculate RAM chips (128 bytes).

Answer: * 8k = 8192 bytes. Decimal range 0 to 8191. * 8191 in Hex is 13 bits of '1': `0001 1111 1111 1111` = **1FFFH**. * Number of chips = $8192 / 128$ = **64 chips**.

Q2. Disadvantage of direct mapping and how to overcome?

Answer: Disadvantage: Conflict Misses (Thrashing). Multiple active memory blocks mapping to the exact same cache line repeatedly evict each other, even if other cache lines are empty. **Overcome:** Use Associative Mapping or Set-Associative Mapping, which allows blocks to map to multiple/any lines.

Q3. OPCODE ADD, operand 206, AC 500, Direct Addressing.

Answer: a) Direct mode means 206 is the actual memory address. AC becomes $500 + M[206]$. b) RTL Specification: $AC \leftarrow AC + M[206]$.

Q4. $X=A+(C^{*}D)$ using zero address instructions.

Answer: Uses a Stack: `PUSH C` `PUSH D` `MUL` (Pops C, D, pushes C*D) `PUSH A` `ADD` (Pops A, C*D, pushes sum) `POP X`

Q5. What are write through and write back policies?

Answer: * **Write-Through:** Every cache write also writes synchronously to main memory. (High memory traffic, but consistent). * **Write-Back:** Modifies only the cache (sets a dirty bit). Writes to main memory only when the block is evicted. (Fast, but temporarily inconsistent).